Tag-Constrained Pseudo-Labels and Two-Decode Voting for Noise Event Detection in Indic Speech

Manish Joshi and Neeraj Singh Aithani
Team Hydra, IndoML Datathon 2026, Track 1

***Abstract*—We detect when non-speech noise events start and end in crowd-sourced Indian speech (Vaani), scored by Combined, the sum of event F1 and segment Dice (maximum 2). Only 12% of the training clips have verified timestamps, and the test set mixes natural recordings with synthetic mixtures. Our key observation is that Vaani transcribers already mark noise events with tags in the transcripts. We use these tags to constrain a teacher ensemble's pseudo-labels, which improves label quality by 0.04–0.05 Combined and labels 305k extra clips. Each model's output is read two ways, by thresholding its activity curve and by an event-proposal head, and a weighted vote over all readouts beats the better single readout by 0.012–0.019 on held-out clips. Our main system is a unified detector: five models trained on balanced natural and synthetic data serve every clip and score 1.6114 on the test set. A routed variant that first separates the organisers' mixtures by their peak level, our final submission, scores 1.6168.**

I. Introduction

Vaani is a crowd-sourced corpus of Indian speech recorded on phones in homes, streets and fields, so clips are full of non-speech noise (horns, animals, babies, music, breathing). The task is to give the onset and offset of every such event, without naming its category. It is hard because few clips have verified timestamps, because the test set mixes natural recordings with synthetic mixtures (speech plus pasted noise) that behave very differently, and because event edges in natural recordings are hard to place: even our best systems reach event F1 of about 0.53 there, against 0.99 on mixtures.

Our starting point is that Vaani transcribers already mark noise events with tags such as `<horn>` … `</horn>`, which say that an event happened and roughly where, but not exactly when. We make three contributions: (i) *tag-constrained pseudo-labels*, which turn these tags into training labels for 305k clips (Sec. II-D); (ii) *two readouts per model and a weighted vote* that fuses them (Sec. III-B); and (iii) a *unified detector* that serves natural and synthetic clips with one set of models (Fig. 1).

II. Training Data and Preprocessing

A. Task and metric. Clips are 16 kHz speech, 1.4–21 s long (median 6.6 s). A predicted event matches a reference event when both its onset and offset lie within $\max(0.2\,d, 0.05)$ s of it (d = reference duration); event F1 counts matches over all events [1]. Segment Dice compares predicted and reference activity on 10 ms frames, averaged over clips; Combined is their sum. The final test set (Phase 2) has 5,517 clips: 3,437 natural and 2,080 synthetic.

B. Data sources. The competition release has three label tiers: 11,111 *Gold* clips with verified timestamps, 61,640 *Silver* clips with unverified timestamps, and 17,884 *Bronze* clips that only list which noise categories occur. We hold out 2,005 Gold clips of distinct speakers as our validation split. The organisers also released 1,806 natural and 695 synthetic labelled validation clips, which we use as training data as well. External data: 321,728 transcribed Vaani clips [2] (tags, no timestamps; 305k kept after labelling), 207,268 untranscribed Vaani clips, the ESC-50 sound-event collection [3], and the noisy/clean pairs of the datathon's speech-enhancement track, used only to train an enhancement network. Every competition clip, and every recording that contains a validation or test clip, was removed from the transcribed pool; the untranscribed pool, used without labels, was filtered clip by clip. No test labels or manual annotation were used; unlabelled test audio was used only to check the router of Sec. III-E and to guide the background of our earliest synthetic mixtures.

C. How each label tier is used. *Gold* is trusted: it gets sharp edge targets and full weight, is sampled three times as often, and trains the boundary refiner (Sec. III-B). *Silver* edges are unreliable, so their targets are blurred, they weigh 0.5–0.9, and their background frames count less; a teacher ensemble moves a Silver edge onto its own event when the two overlap (IoU $\geq$0.3, shift $\leq$0.5 s) and adds events it agrees on strongly where Silver has none. *Bronze* has no timestamps: it supervises only a clip-level loss until the tag-constrained labels below give it timestamps (17,373 clips with transcripts), at weight 0.35–0.5. The *transcribed* pool is labelled the same way, and the *untranscribed* pool serves the pretext task (Sec. III-C) and early teacher labels where the ensemble agreed.

D. Tag-constrained pseudo-labels. A *teacher* is a set of trained models whose readouts are fused into candidate events, each with a vote count (its *support*). For a transcribed clip we keep the event tags (not `<noise>`, `<PAUSE>` or `[unintelligible]`) and turn each into a time prior: a tag occupying the fraction $[a, b]$ of the tag-free text predicts an event centred at $t_{\text{tag}} = \frac{a+b}{2}\,d$. Hungarian matching assigns each tag exactly one candidate centred at t_{cand}, minimising $3\,|t_{\text{cand}} - t_{\text{tag}}|/d - v/v_{\max}$ (time distance against support v, with $v_{\max}$ the number of readouts); unmatched candidates are dropped (Fig. 2a). On Gold clips the tag count equals

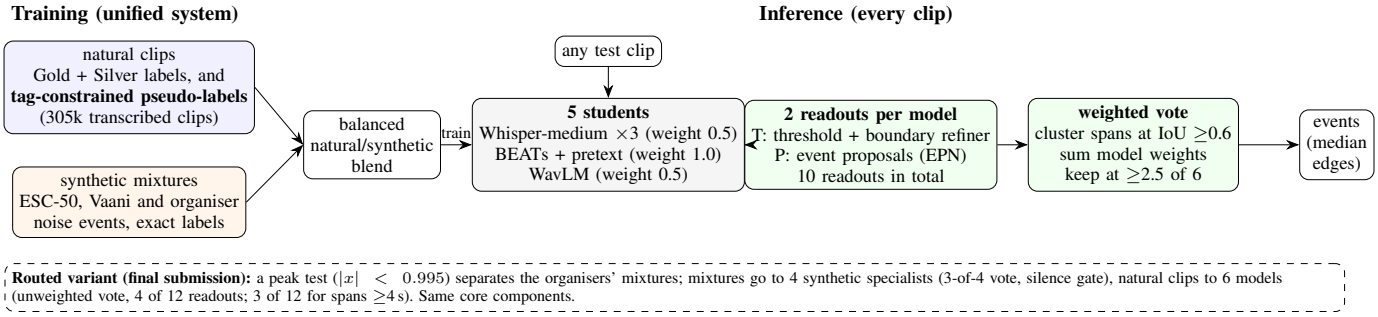


Fig. 1: The unified system, our main method: one balanced blend trains five students, and every test clip, natural or synthetic, goes through the same students, two readouts each and one weighted vote. Dashed: the routed variant.

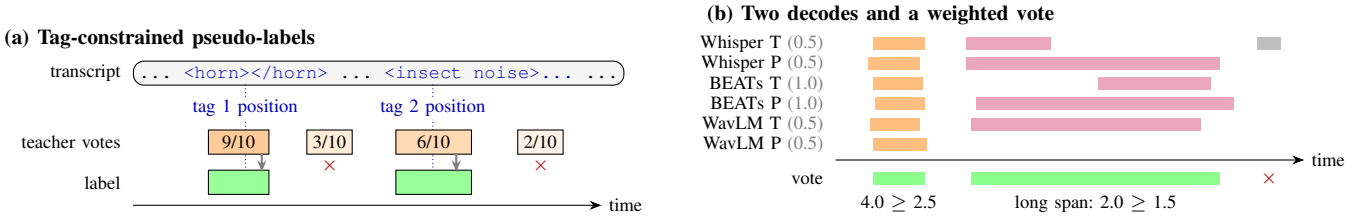


Fig. 2: The two core ideas. (a) Tags fix how many teacher candidates become labels; Hungarian matching on time and support picks them (votes out of 10 teacher decodes). (b) T = threshold readout with refiner, P = proposal readout; three of the five models shown. The short event has all six readouts ($4.0 \geq 2.5$); the long one only Whisper P, BEATs P and WavLM T (2.0), kept by the long-span rule (1.5; used by the routed system, optional for the unified one); the isolated span (0.5) is dropped. Not to scale.

the verified event count 95.8% of the time, so the count says how many events to keep and the positions say which. Scored against our held-out Gold clips, the constraint lifts a three-model teacher's labels from 1.1903 to 1.2415 Combined and the five-model teacher's, used for the final models, from 1.2145 to 1.2578.

E. Synthetic mixtures. To cover the synthetic half of the test set we build mixtures like the organisers': up to six ESC-50, Vaani or organiser noise events (2,213, recovered from their synthetic validation clips by subtracting the clean references) pasted into event-free stretches of Gold clips, then peak-normalised.

F. Preprocessing and augmentation. Audio is 16 kHz mono, turned into 128-bin log-mel features on a 10 ms grid, and models train on random crops of up to 15 s. Augmentation is the Gold oversampling above, mixup ($\alpha = 0.2$, union of labels) for the BEATs and Dasheng synthetic specialists (Sec. III-E), and, for three BEATs models, a consistency loss between two views of a clip that differ by a time shift of up to 0.2 s, a gain change of up to 6 dB and added noise. The Whisper and WavLM models use crops and Gold oversampling only; the synthetic mixtures of Sec. II-E act as the main domain augmentation, and speed, gain and equaliser augmentation are off in all final models.

III. Proposed Method

A. Base model. Every model is a convolutional recurrent network [4]: convolutions over the log-mel features are combined with the frames of one pretrained encoder (Table I), and a bidirectional GRU predicts, per 10 ms frame, event activity, onset and offset likelihoods and, through an Event Proposal Network (EPN) [5], the distance to the start and end of the current event. The loss combines frame and boundary cross-entropy with an IoU loss on the proposals (weight 2), weighted by tier as in Sec. II-C. We train with AdamW (weight decay 0.01), warmup and cosine decay, and an exponential moving average of the weights (0.999); learning rates are 10^{-4} for full training and 3×10^{-5} for students, with the encoder at a tenth of that. Each final model ends a short chain: a model with a frozen encoder, its fine-tuned copy, then a one-epoch *student* trained on teacher labels [6], [7]; earlier stages run 2–6 epochs.

B. Two readouts and the vote. A model's frame outputs are read out twice. The threshold readout thresholds and smooths the activity curve, then a small boundary refiner, trained on Gold edges displaced by up to ±250 ms, moves each edge to its most likely true position; the proposal readout takes whole events from the EPN. The two make different errors, so voting over both helps (Table III). The vote clusters overlapping spans (IoU ≥0.6), adds each readout's model weight, and keeps a cluster whose summed weight passes a threshold, with median edges (Fig. 2b). A *long-span rule* lets long clusters, which single readouts often split, pass with less weight.

C. Localisation pretext. General audio encoders recognise sounds but were not trained to time them, so before fine-tuning BEATs and Dasheng spend 10k steps finding 0–3 ESC-50 sounds pasted into untranscribed Vaani audio; the share of pasted events they localise within the competition's tolerance rises from 0.059 to 0.706 (BEATs) and 0.728 (Dasheng).

D. Unified system (our main method). Five students (three Whisper-medium, one BEATs with the pretext, one WavLM)

TABLE I: Pretrained models: version, source, and how many final models of each system use them as encoder.

Model	Version, source	Role
Whisper [8]	medium; OpenAI	4 routed, 3 unified
WavLM [9]	base-plus; Microsoft	1 routed, 1 unified
HuBERT [10]	large; Meta	initialised WavLM models
BEATs [11]	iter3+ AS2M; Microsoft	3 routed, 1 unified
Dasheng [12]	base, v0.0.9; Xiaomi	2 routed
Enhancement U-Net	ours (enhancement track)	noise input, 1 routed

TABLE II: Phase 2 test set (5,517 clips). Natural / Synthetic give event F1 / Dice on each subset; – = not logged. †Both derived: overall Dice is the clip-weighted mean of the subset Dice (exact for our routed entry); F1 = Combined − Dice.

System	Combined	F1 / Dice	Natural	Synthetic
Ours, earlier single model	1.5883	–	.484 / .708	.996 / .992
Ours, routed	**1.6168**	.790 / .827	.526 / .726	.997 / .992
Ours, unified	**1.6114**	.793 / .819†	.526 / .715	.993 / .990

are trained on roughly equal shares of natural clips with tag-constrained labels and synthetic mixtures with exact labels. One vote fuses their ten readouts on every clip (Whisper and WavLM weigh 0.5, BEATs 1.0; keep at 2.5 of 6), with no knowledge of the clip's type (Fig. 1).

E. Routed variant (our final submission). The organisers' mixtures are peak-normalised to 0.99 while natural clips reach full scale, so a peak test ($|x| < 0.995$) separates them exactly; this exploits a property of the release, not of the audio. Mixtures go to four synthetic specialists (BEATs, two Dasheng, Whisper; the BEATs one also sees the enhancement network's noise estimate as a second input), followed by a gate that drops spans more than 30 dB below the clip's loudest frames; natural clips go to six models (three Whisper, two BEATs, one WavLM) with the long-span rule from 4 s (Fig. 1, dashed).

IV. Results and Conclusion

Test-set scores. Our routed system scores 1.6168 (event F1 0.790, Dice 0.827) on the Phase 2 test set and the unified system 1.6114, against 1.5883 for one of our earlier single models (Table II); retraining one model with another seed moves the score by up to ±0.002, so smaller differences are not meaningful. The unified system has the higher event F1 (0.793, derived), so its gap is Dice, the frame overlap, driven by edge placement.

What each idea contributes. In Table III, tag positions, not counts, carry the label gain for both teachers, and a student trained on tag-constrained labels slightly improves on one trained on plain labels from the same audio (+0.0036, one seed). On held-out clips, voting over five models adds 0.036, fusing both readouts of each model a further 0.012–0.019 over the better single readout, and the long-span rule 0.004–0.006; the pretext task helps on the organisers' clips only.

Where natural clips fail. For an earlier ensemble on the organisers' natural clips, events under 0.25 s matched 13% of the time and 47% of events over 4 s were under half covered,

TABLE III: Evidence per idea. L = label quality vs held-out Gold, V = our validation, O = organisers' natural clips. †Five released unified-system models, none trained on V; one saw 2 of the 1,806 O clips. ‡95% speaker-bootstrap interval above zero.

Comparison (A vs B)	On	Δ Combined
Tag-constrained pseudo-labels		
Tags vs none, 3-model teacher	L	+0.051
Tags vs none, 5-model teacher	L	+0.043
Tag count only vs none, 3-model teacher	L	+0.003
Student on tag vs plain labels (1 seed)	V	+0.0036
Two readouts and the vote (columns: V / O)		
†Vote of 5 models vs 1 model (proposal)	V / O	+0.036 / +0.036
†‡Both readouts vs proposal-only vote	V / O	+0.019 / +0.012
†Both readouts vs threshold-only vote	V / O	+0.028 / +0.028
†‡Long-span rule vs none	V / O	+0.006 / +0.004
Other components		
Pretext task vs none (1 BEATs)	V / O	−0.003 / +0.017

while only 127 of 2,531 predictions touched no event: the error is edge placement, not detection.

Reproducibility and disclosures. Code and configurations are at https://github.com/mcjoshi/indic-noise-event-detection, checkpoints at https://huggingface.co/manishjoshiai/indic-noise-event-detection; the released code reproduces both submissions, except one clip that the submitted routed file had filled from an earlier submission. Late choices were made on the test leaderboard, so leaderboard gains are optimistic; four natural models train on the organisers' natural clips, and all synthetic and unified models on their synthetic clips.

Conclusion. Transcript tags are a free way to constrain pseudo-labels, and voting over two readouts per model gives one detector for both kinds of clip; placing edges in natural recordings remains the open problem.

References

[1] A. Mesaros, T. Heittola, and T. Virtanen, "Metrics for polyphonic sound event detection," *Applied Sciences*, 2016.

[2] ARTPARK, IISc and Google, "Project Vaani," https://vaani.iisc.ac.in, 2025.

[3] K. J. Piczak, "ESC: Dataset for environmental sound classification," in *Proc. ACM Multimedia*, 2015.

[4] E. Çakır *et al.*, "Convolutional recurrent neural networks for polyphonic sound event detection," *IEEE/ACM TASLP*, 2017.

[5] F. Schmid, C. I. Tang, S. Parekh, and V. K. Ithapu, "Sound event detection with boundary-aware optimization and inference," arXiv:2601.04178, 2026.

[6] Q. Xie, M.-T. Luong, E. Hovy, and Q. V. Le, "Self-training with noisy student improves ImageNet classification," in *CVPR*, 2020.

[7] F. Schmid *et al.*, "Improving audio spectrogram transformers for sound event detection through multi-stage training," arXiv:2408.00791, 2024.

[8] A. Radford *et al.*, "Robust speech recognition via large-scale weak supervision," in *ICML*, 2023.

[9] S. Chen *et al.*, "WavLM: Large-scale self-supervised pre-training for full stack speech processing," *IEEE JSTSP*, 2022.

[10] W.-N. Hsu *et al.*, "HuBERT: Self-supervised speech representation learning by masked prediction of hidden units," *IEEE/ACM TASLP*, 2021.

[11] S. Chen *et al.*, "BEATs: Audio pre-training with acoustic tokenizers," in *ICML*, 2023.

[12] H. Dinkel *et al.*, "Scaling up masked audio encoder learning for general audio classification," in *Interspeech*, 2024.